

Exploring "What-If" Scenarios Using a Leslie Matrix

The purpose of this assignment is to use an age-structured Leslie matrix model to investigate how changes in fecundity and survival affect the long-term dynamics of a population. You will compare several hypothetical management scenarios, evaluate their effects on the dominant eigenvalue and stable age distribution, and interpret the biological meaning of the results.

Background

A Leslie matrix has the form

$$L = \begin{pmatrix} F_1 & F_2 & \cdots & F_n \\ S_1 & 0 & \cdots & 0 \\ 0 & S_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & S_{n-1} \end{pmatrix},$$

where F_i denotes the fecundity of age class i and S_i denotes the probability of surviving from age class i to age class $i + 1$.

The dominant eigenvalue, λ , describes the asymptotic population growth rate:

$\lambda > 1$ population growth,

$\lambda = 1$ stable population,

$\lambda < 1$ population decline.

The normalized right eigenvector associated with λ gives the stable age distribution.

Learning Objectives

After completing this assignment, you should be able to

- use a Leslie matrix to represent an age-structured population;
- compute the dominant eigenvalue and stable age distribution;
- modify fecundity and survival parameters to create management scenarios;
- compare scenario outcomes with a baseline model;
- explain the limitations of scenario analysis.

1 Baseline Model

1. Enter the baseline Leslie matrix in Python.
2. Compute all eigenvalues and identify the dominant eigenvalue λ .
3. Compute the corresponding stable age distribution.
4. Verify that the entries of the stable age distribution sum to 1.
5. Interpret the biological meaning of λ .

A possible code structure is shown below.

```
import numpy as np

L = np.array([
    # Enter the Leslie matrix here
])

eigenvalues, eigenvectors = np.linalg.eig(L)

# TODO:
# 1. Identify the dominant eigenvalue.
# 2. Extract the corresponding right eigenvector.
# 3. Normalize the eigenvector so its entries sum to 1.
```

2 Single-Parameter Scenarios

For each scenario below, modify only one parameter while keeping all other entries of the Leslie matrix unchanged.

Fecundity Scenarios

Decrease each nonzero fecundity parameter by 10%, one at a time.

For each modified matrix,

1. Compute the new dominant eigenvalue;
2. Compute the new stable age distribution;
3. Calculate the change $\Delta\lambda = \lambda_{\text{scenario}} - \lambda$;
4. Calculate the Euclidean distance between the scenario stable age distribution and the baseline stable age distribution: $d = \|\mathbf{w}_{\text{scenario}} - \mathbf{w}\|_2$.

Survival Scenarios

Repeat the same analysis after decreasing each nonzero survival probability by 10%, one at a time.

3 Summary Table

Create a table containing the following columns:

Scenario	Modified parameter	Dominant eigenvalue	SAD distance
Baseline	None	λ	0
Fecundity scenario 1	F_i decreased by 10%		
Survival scenario 1	S_i decreased by 10%		

Rank the scenarios from the largest decrease in λ to the smallest.

4 Visualization

Produce the following figures.

Figure 1: Stable Age Distributions

Plot the baseline stable age distribution together with the stable age distributions produced by the single-parameter scenarios. The figure should include age class on the horizontal axis, proportion on the vertical axis, distinct lines or markers for each scenario, and a readable legend.

Figure 2: Scenario Effects

Create a second figure comparing $\Delta\lambda$ for each scenario; the distance between each scenario stable age distribution and the baseline. Choose a plot type that makes comparison among scenarios clear.

5 Hypothetical Management Scenarios

Consider a hypothetical management scenario. Suggested examples are

Scenario A: Increase juvenile survival by 10%.

Scenario B: Increase adult survival by 10%.

Scenario C: Increase fecundity of older age classes by 10%.

Scenario D: Increase both juvenile survival and adult fecundity by 5%.

Scenario E: Design your own biologically motivated management scenario.

For the chosen scenario, define the modified Leslie matrix, compute its dominant eigenvalue and stable age distribution, and compare the results with those of the baseline model.

Interpretation

Answer the following questions in complete sentences.

1. Which single-parameter change produced the largest decrease in the dominant eigenvalue?
2. Which single-parameter change altered the stable age distribution the most?
3. Did the same parameter produce the largest change in both measures?
4. Which age classes appear to be most important for long-term population growth?
5. How do your conclusions compare with the sensitivity and elasticity analysis?
6. What assumptions of the Leslie matrix should be considered before making management recommendations?

Deliverables

1. A completed Jupyter notebook containing all required code;
2. A summary table comparing the baseline model with the selected scenarios;
3. All required figures;
4. A one- to two-page report interpreting the results.